Korea Technology and Information Promotion Agency for SMEs

*(max 10 pages)*

# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information


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## 2. Technology Overview

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| Title of Proposed Technology | Advanced Dielectric Nanofiller Technology for Directional Heat Dissipation in Semiconductor Packaging Materials | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1)] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | Electrical · Electronics | | Chemicals | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Semiconductor Devices and Systems | Semiconductor Materials | Polymeric Materials | Nanomaterial Technologies |
| Concept of Proposed Technology | The core of the proposed technology lies in the implementation of a dual design strategy for nanofiller-based composite materials that simultaneously maintain electrical insulation while exhibiting high thermal conductivity, by (i) fundamentally reducing interfacial thermal resistance, and (ii) controlling the directionality of heat transport pathways.<br>Conventional thermal management materials for semiconductor packaging have primarily relied on increasing filler loading to enhance thermal conductivity.[1-3] However, this approach inevitably introduces critical limitations, including increased interfacial thermal resistance, non-uniform filler dispersion, and degradation of electrical properties.[4-7] In packaging materials that require strict electrical insulation, the use of conductive fillers is not permitted, rendering interfacial thermal resistance in dielectric nanofiller-based systems a key technological bottleneck. In addition, a mismatch between the direction of heat generation and the desired heat dissipation pathway often leads to inefficient, undirected heat spreading.[8,9]<br>To overcome these challenges, the proposed technology integratively realizes: (i) Composite material design employing electrically insulating yet highly thermally conductive nanofillers, (ii) Directional control of heat transport pathways through spatial organization and self-assembly of nanofillers, and<br>(iii) reduction of interfacial thermal resistance via nanofiller–polymer interfacial engineering.<br>Accordingly, the proposed composite material is not merely a "highly thermally conductive material," but rather a directional thermal management insulating composite capable of guiding and dissipating heat along predefined pathways under suppressed interfacial thermal resistance. This | | | |

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| | integrated capability fundamentally differentiates the proposed technology from conventional thermal management materials. |
| **Rationale for Joint R&D Project** | In advanced semiconductor packaging, performance requirements extend beyond the achievement of high thermal conductivity alone and increasingly demand the simultaneous maintenance of electrical insulation, control over the directionality of heat transport, and compatibility with industrial manufacturing processes.[10,11] In particular, in highly integrated and high-power devices, heat is generated in a localized manner; if this heat is not effectively guided toward external heat-dissipation pathways, package reliability and operational lifetime can deteriorate rapidly.[12]<br><br>Meeting these stringent requirements necessitates the integrated combination of:<br>• Nanofiller materials that simultaneously exhibit electrical insulation and high thermal conductivity,<br>• Design of heat transport pathways through directional alignment and structural control of nanofillers,<br>• Minimization of heat transfer losses at nanofiller–polymer interfaces, and<br>• Experience in composite material fabrication and application at an industrial process level.<br><br>To address these multifaceted demands, the proposed international joint R&D project aims to synergistically integrate:<br>• the fundamental technological capabilities of a research institution specializing in nanoparticle synthesis, interfacial engineering, nanoscale self-assembly, and directional structure control[13-15], and<br>• the industrial expertise of a Korean SME with hands-on experience in developing insulating, high-thermal-conductivity nanofiller-based composites at the process level and optimizing them for semiconductor packaging applications.<br><br>Through this complementary collaboration, the project seeks to ensure that the concept of directional heat transport advances beyond laboratory-scale demonstrations and is translated into practically deployable composite material technologies for advanced semiconductor packaging. |
| **Key Words** | 1. Nanofiller interface engineering<br>2. Thermally conductive dielectric composites<br>3. Heat dissipation control<br>4. Advanced semiconductor packaging |

## 3. Research and Development Objectives & Plan

| **Technology Readiness Level** (TRL)[2)] | **Start of the Project** | Level (2-3) | **End of the Project** | Level (4-5) |
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| **R&D Objectives** | The objective of this R&D project is to establish a dielectric nanofiller composite platform that enables efficient and directional heat dissipation in semiconductor packaging applications.[16-20] By overcoming interfacial thermal resistance limitations inherent to conventional filler-loading strategies, the project aims to deliver a new class of electrically insulating thermal management materials capable of addressing localized heat-generation challenges in advanced semiconductor devices.<br>Specifically, this project seeks to (i) define interfacial design principles that enhance thermal transport across nanofiller–polymer interfaces, (ii) develop composite fabrication strategies that promote directional heat-flow pathways, and (iii) validate composite-level thermal and dielectric performance under semiconductor-relevant conditions.[21-25] Collectively, these efforts will establish a transferable materials design framework applicable to next-generation semiconductor packaging technologies.[26-28] | | | |

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| **R&D Target Specifications** | The final goal of this project is to demonstrate a dielectric nanofiller composite material that meets the following target specifications under laboratory-scale, application-relevant conditions:<br><br>1. Composite Thermal Conductivity[10,29]<br>Target: ≥ 1.2 W/m·K<br>This specification reflects the minimum effective thermal conductivity required to meaningfully enhance heat dissipation in advanced semiconductor packaging while remaining compatible with electrically insulating polymer matrices. Achieving this level of performance through interfacial engineering rather than excessive filler loading is critical for preserving mechanical integrity and processability.<br><br>2. Interfacial Thermal Resistance[30,31]<br>Target: ≤ $1.5 \times 10^{-8}$ $m^2 \cdot K/W$<br>Reduction of interfacial thermal resistance is a central technical objective of this project, as interfacial losses dominate heat transport in dielectric nanofiller-based composites. This target value is selected to represent a substantial improvement over conventional untreated nanofiller–polymer systems, enabling more efficient phonon transport across interfaces.<br><br>3. Dielectric Performance[32]<br>Target: Dielectric constant ≤ 4.0 (at 1 MHz)<br>Maintaining low dielectric constant is essential for semiconductor packaging applications. This specification ensures that enhancements in thermal performance do not compromise electrical insulation, signal integrity, or device reliability. |
| **R&D Plan** | **Aim 1. Joint Establishment of an Integrated Interfacial Thermal Engineering and Directional Control Design Framework** (0–6 months)<br><br>The objective of this aim is to establish a common design framework for dielectric nanofiller composites that integrates interfacial thermal engineering with nanofiller directional control. In the initial phase, close technical coordination between the research institute and the Korean SME will ensure alignment of material design concepts, processing constraints, and semiconductor packaging requirements from the outset of the project.<br>Representative semiconductor packaging architectures, localized heat-generation geometries, and target heat dissipation pathways will be jointly defined. Based on this application context, candidate nanofiller systems, polymer matrix compatibility, and feasible processing routes will be coordinated. Through this process, design requirements for reducing interfacial thermal resistance and enabling directional heat transport will be clearly defined, establishing the foundation for subsequent interfacial modification and structural control efforts.<br><br>***Milestones:*** (i) Joint selection of nanofiller–polymer systems suitable for semiconductor packaging; (ii) Definition of design requirements for interfacial thermal resistance reduction and directional heat transport; (iii) Establishment of an integrated interfacial and directional design framework<br><br>**Aim 2. Development of Nanofiller Surface Modification Strategies for Interfacial Thermal Resistance Reduction and Interfacial Validation** (6–16 months)<br><br>The objective of this aim is to systematically develop nanofiller surface modification strategies that effectively reduce thermal resistance at nanofiller–polymer interfaces and to validate their impact at the interfacial level. This aim focuses on surface chemistry–based approaches that enhance interfacial thermal coupling while maintaining dielectric properties.<br>Surface modification strategies will be designed and implemented to improve interfacial thermal transport, while simultaneously evaluating nanofiller dispersion stability and process compatibility. The developed surface modification approaches will be assessed using model composite systems to quantitatively verify interfacial thermal resistance reduction and interfacial stability, thereby establishing a robust physical and chemical basis for subsequent composite structural design and performance evaluation.<br><br>***Milestones:*** (i) Establishment of nanofiller surface modification strategies for interfacial thermal |

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| | resistance reduction; (ii) Validation of dispersion stability and interfacial characteristics of surface-modified nanofillers; (iii) Quantitative confirmation of interfacial thermal resistance reduction<br><br>**Aim 3. Composite Structural Design Based on Nanofiller Orientation Control and Validation of Directional Heat Dissipation** (14–26 months)<br><br>The objective of this aim is to implement composite architectures that enable controlled orientation and alignment of nanofillers within polymer matrices, and to experimentally validate their impact on directional heat dissipation. Surface-modified nanofillers developed in Aim 2 will be incorporated to ensure stable interfacial thermal coupling.<br>In this aim, nanofiller orientation will be utilized as a structural design parameter to support directional heat dissipation that is consistent with heat dissipation pathways defined by semiconductor packaging architectures. The fabricated composites will be evaluated to establish quantitative relationships between nanofiller orientation, composite structure, and directional thermal transport behavior.<br><br>***Milestones:*** (i) Implementation of composite structures enabling nanofiller orientation control; (ii) Experimental validation of directional heat dissipation behavior; (iii) Establishment of correlations between nanofiller orientation and thermal transport performance<br><br>**Aim 4. Integrated Thermal and Dielectric Performance Validation and Application-Oriented Design Refinement** (24–36 months)<br><br>The objective of this aim is to perform integrated validation of thermal and electrical performance of the interfacial- and orientation-engineered composites under semiconductor packaging–relevant conditions, and to refine composite designs with consideration of practical application and scalability.<br>Thermal conductivity, directional heat dissipation behavior, and dielectric properties will be evaluated in a comprehensive manner to verify compliance with target performance specifications. Process compatibility, manufacturing reproducibility, and scalability considerations will be assessed to establish application-ready composite design guidelines at the TRL 4–5 level.<br><br>***Milestones:*** (i) Integrated validation of thermal and dielectric performance under semiconductor-relevant conditions; (ii) Assessment of process compatibility and manufacturing reproducibility; (iii) Definition of application-oriented composite design guidelines at TRL 4–5 |

## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | In this project, Rensselaer Polytechnic Institute (RPI) will focus on technical elements related to nanofiller–polymer interfacial design and analysis of interfacial thermal transport behavior. Major activities include the design of nanofiller–polymer interfacial architectures, development of nanofiller surface modification strategies to reduce interfacial thermal resistance, and fabrication of dielectric nanofiller-based composites with consideration of semiconductor packaging applications.<br><br>Formulation-level screening and optimization will be conducted to evaluate nanofiller dispersion stability and composite processability. In parallel, systematic analysis will be performed to elucidate relationships among nanofiller surface chemistry, interfacial thermal resistance, and thermal transport behavior while maintaining electrical insulation properties.<br><br>RPI will also carry out thermal and electrical performance evaluation of the fabricated composites to deepen understanding of interfacial thermal transport mechanisms, and to derive processing guidelines and interfacial design principles applicable to semiconductor packaging materials. The resulting technical know-how—including surface modification protocols, composite fabrication methodologies, and performance evaluation frameworks—will be organized to support industrial application and follow-on development by the Korean SME partner. |

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| **Expected R&R of Korean SME Partner** | The Korean SME partner will leverage prior experience in dielectric, thermally conductive composite materials based on nanomaterial fillers and their application to semiconductor packaging processes to define application-driven performance targets, material design scope, and relevant industrial constraints within this project.<br><br>Key activities include deriving technical requirements related to nanomaterial-filled composite design and process integration, assessing scalability and manufacturing feasibility, and evaluating the industrial relevance and commercialization potential of the developed dielectric nanofiller and composite concepts. Through these activities, the Korean SME partner will provide application-oriented technical feedback to ensure that the proposed interfacial designs and composite architectures are compatible with practical semiconductor packaging environments.<br><br>In addition, the Korean SME partner will support application-oriented validation and commercialization planning, including assessment of compatibility with existing semiconductor packaging workflows, consideration of cost and manufacturing factors, and analysis of potential product integration pathways. Through close collaboration with RPI, the Korean SME partner will facilitate technology transfer, scale-up strategy development, and downstream industrial adoption and commercialization of the developed dielectric nanofiller-based composite technologies. |

## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | This joint R&D project aims to deliver both technological advancements and commercialization-oriented outcomes in dielectric thermal management materials for advanced semiconductor packaging applications.<br><br>From a technical perspective, the project will establish interfacial thermal engineering strategies that effectively reduce interfacial thermal resistance at nanofiller–polymer interfaces, while integrating design concepts that enable controlled orientation and alignment of dielectric nanofillers within polymer matrices. By combining nanofiller surface modification with nanofiller orientation control, the project seeks to enhance heat dissipation performance in dielectric nanofiller-based composite materials while maintaining electrical insulation. In this context, controlled nanofiller orientation may serve as a design element that supports directional heat dissipation consistent with heat dissipation pathways defined by semiconductor packaging architectures. Through the application of facile and scalable nanofiller surface modification techniques, the project will generate quantitative structure–property relationships linking nanofiller surface chemistry, nanofiller–polymer interfacial thermal resistance, nanofiller orientation, and composite-level thermal transport performance. These outcomes will provide transferable interfacial design principles and processing guidelines applicable across a wide range of dielectric nanofiller systems relevant to semiconductor packaging.<br><br>From an industrial and commercialization perspective, the project will establish a foundation for translating the nanofiller surface modification technologies and composite fabrication strategies developed in this work into practical material platforms for the Korean SME partner. The surface-modified nanofillers and composite fabrication protocols generated through this project are expected to be compatible with existing semiconductor packaging processes, thereby facilitating technology transfer, scale-up, and downstream product development. Through this collaboration, the Korean SME is expected to strengthen its technological competitiveness by delivering differentiated thermal management solutions that extend beyond conventional filler-loading approaches and leverage nanofiller orientation as an additional design parameter.<br><br>In a broader context, successful completion of this project will contribute to improved thermal reliability and performance of high-density and high-power semiconductor packages, helping to alleviate critical thermal bottlenecks in next-generation electronics. The outcomes are expected to support global market expansion of Korean SMEs in advanced packaging materials, while reinforcing the role of international collaborative R&D as an effective pathway for accelerating innovation, commercialization, and industrial impact in strategically important technology areas. |

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## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
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| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |